Enabling Few-Shot Alzheimer's Disease Diagnosis on Tabular Biomarker Data with LLMs

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Abstract

Early and accurate diagnosis of Alzheimer's disease (AD), a complex neurodegenerative disorder, requires analysis of heterogeneous biomarkers (e.g., neuroimaging, genetic risk factors, cognitive tests, and cerebrospinal fluid proteins) typically represented in a tabular format. With flexible few-shot reasoning, multimodal integration, and natural-language-based interpretability, large language models (LLMs) offer unprecedented opportunities for prediction with structured biomedical data. We propose a novel framework called **TAP-GPT**, *Tabular Alzheimer's Prediction GPT*, that adapts TableGPT2, a multimodal tabular-specialized LLM originally developed for business intelligence tasks, for AD diagnosis using structured biomarker data with small sample sizes. Our approach constructs few-shot tabular prompts using in-context learning examples from structured biomedical data and finetunes TableGPT2 using the parameter-efficient qLoRA adaption for a clinical binary classification task of AD or cognitively normal (CN). The TAP-GPT framework harnesses the powerful tabular understanding ability of TableGPT2 and the encoded prior knowledge of LLMs to outperform more advanced general-purpose LLMs and a tabular foundation model (TFM) developed for prediction tasks. To our knowledge, this is the first application of LLMs to the prediction task using tabular biomarker data, paving the way for future LLM-driven multi-agent frameworks in biomedical informatics.

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CCS Concepts

• **Computer systems organization** → *Neural networks*; • **Applied computing** → **Health informatics**; • **Computing methodologies** → *Transfer learning*; **Model development and analysis**; **Natural language processing**.

Keywords

Alzheimer's Disease Biomarkers, Tabular Prediction, Large Language Models, AI and Machine Learning in Healthcare

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1 Introduction

Alzheimer's disease (AD) is a neurodegenerative disease with a complex etiology and is the sixth leading cause of death in the United States. Globally, AD is the single most common form of dementia and accounts for over two-thirds of all 55 million dementia cases, according to World Health Organization[1]; and the projected number of AD and AD-related dementia cases could surge to more than 100 million by 2050 [20, 21]. AD, to date, is highly prevalent and has no effective pharmacologic treatment alternatives for a cure; moreover, post-treatment relapse remains stubbornly elevated. Recently, an increasing number of studies have demonstrated that early detection and diagnosis could result in proactive treatment and a significant slow-down in AD progression. Diverse AD biomarkers (e.g., neuroimaging, genetics, cognitive tests, fluid biomarkers) offer unprecedented opportunities for early diagnosis [12, 13]. For example, Quantitative templates for the progression of AD (QT-PAD)[2], the AD Modeling Challenge from ADNI, gathers a list of 16 clinical, cognitive, and imaging biomarkers, such as cerebrospinal fluid

[1]https://www.who.int/news-room/fact-sheets/detail/dementia
[2]www.pi4cs.org/qt-pad-challenge

(CSF) tau protein level, whole brain volume, as well as age, gender, education covariates. However, unlike text or imaging data, such biomarker data are naturally stored in a table format with columns representing each marker and rows representing each subject. The table often comprises of heterogeneous data types and is also missing order or locality information, which presents unique challenges for developing predictive models compared to the rich information provided in text or image data [6, 18].

Most existing AI models developed to handle AD biomarkers can be categorized into machine learning or classic deep neural networks with limited knowledge of understanding AD [2, 17, 22]. Traditional machine learning pipelines, e.g. ensembles of decision trees, have remained dominant for tabular biomedical data in general because they are robust to small data, whereas deep learning models often struggle to capture such tabular patterns [6, 18]. However, these conventional models offer limited flexibility in combining modalities and do not leverage any prior biomedical knowledge beyond the training data. This motivates exploration of foundation models for tabular data that can perform few-shot reasoning and incorporate broad prior knowledge, potentially improving prediction from biomedical tables. The recent advancements of Transformer-based large language models open up new opportunities for handling tabular data [1, 5]. pretrained on extensive text corpora, LLMs exhibit emergent and versatile abilities in a range of tasks related to tabular data, including tabular data generation, table understanding, tabular prediction, and reasoning, *etc.* based on recent studies [7, 8, 14, 16, 25, 34]. Especially, studies suggest that large language models can be repurposed for tabular tasks by appropriate prompting or finetuning. Unlike vanilla transformers trained purely on text, which cannot directly input large tables or generalize well to tabular formats, new frameworks have emerged to bridge this gap. TableGPT [31] introduced a unified approach combining tables with natural language and functional commands, enabling GPT-style decoders to interpret and manipulate tables through natural language instructions. It developed a dedicated table encoder that lets an LLM access an entire table as a compact vector, overcoming token-length limitations and capturing schema-level context. Building on top of this, TableGPT2 [23] scaled up with pretraining on 593K tables and 2.36M table-question examples, and a novel schema-aware table encoder integrated into a 7B–72B parameter QWen2.5 decoder [11]. This large-scale multimodal LLM achieves remarkable accuracy gains (∼35–49% improvement over base LLMs) on several table-related business intelligence tasks, confirming the value of specialized tabular pretraining. Meanwhile, benchmarks like TableBench [30] evaluated LLM reasoning on complex Table QA tasks and found that LLMs still perform worse than human-level on challenging multi-step reasoning with real-world tables. These studies highlight both the promise of LLMs for structured data and the remaining gap in fully capturing real-world tabular complexity.

Moreover, beyond LLM-based approaches, tabular foundation models specialized for small to medium-sized datasets have been developed. Notably, TabPFN [9, 10] is a transformer model that performs in-context learning (ICL) on tabular inputs: given a small training set in a table, it instantly predicts labels for new samples without any gradient updates. ICL first emerged in large language models [3], yet later studies revealed that transformers can use ICL to approximate machine learning algorithms such as logistic regression and neural networks [19, 35]. TabPFN's single forward-pass classification, enabled by prior-fitting on 100 million synthetic datasets encoding diverse causal structures, yields impressive accuracy on par with state-of-the-art methods. In fact, TabPFN can often outperform gradient-boosted tree ensembles on small tables with many outliers or missing values, needing as little as 50% of the data to match the accuracy of conventional models. Subsequent work has extended this paradigm: a retrieval-augmented finetuning strategy was proposed to adapt TabPFN to larger or more complex tasks by fetching nearest-neighbor subsets of data and finetuning the model on them [24]. This approach achieved impressive results on a 95-dataset benchmark, even outperforming tuned XGBoost models and substantially boosting TabPFN's base performance. Meanwhile, researchers have also explored lighter-weight alternatives to full model training. For example, Wu and Hou [29] introduced an efficient retrieval-based method that uses a frozen LLM's embedding capabilities to perform k-nearest-neighbor prediction in the LLM latent space. By avoiding any model finetuning and leveraging only the target dataset and the LLM's pretrained knowledge, their method achieved surprisingly strong tabular prediction performance. These progress underscore a growing consensus that both LLMs and specialized tabular transformers can be harnessed for table data tasks through careful design.

However, none of the above approaches have been applied to heterogeneous biomedical tabular data such as AD biomarkers, which pose unique challenges of multimodal feature integration and often times limited sample sizes (e.g. typically only a few hundred patients in research cohorts). Moreover, since interpretability is critical in clinical settings, LLMs may offer advantages by producing natural language rationales for their predictions. Here, we aim to fill this gap by leveraging the recent tabular-focused LLM advancements for AD diagnosis. In this work, we propose *TAP-GPT: Tabular Alzheimer's Prediction GPT*, a novel framework that adapts the TableGPT2 multimodal LLM to the task of classifying AD vs cognitively normal (CN) individuals from biomarker tables. We formulate the problem in a tabular few-shot in-context learning paradigm: the model is prompted with a small number of example subjects with biomarkers (including features from neuroimaging, genotyping, etc.) along with their diagnoses, and asked to predict the diagnosis for a new subject, all presented in a single table. To adapt TableGPT2 to this completely new task, we apply a parameter-efficient finetuning (qLoRA) [4] using a small training set available. The resulting TAP-GPT framework combines the powerful table comprehension and reasoning abilities of TableGPT2 with prior medical knowledge encoded in the backbone LLM's weights. Our experiments show that TAP-GPT achieves promising performance, compared to both general-purpose LLMs (which lack tabular specialization) and the tabular foundation model TabPFN on the AD classification task. This work represents, to our knowledge, the first application of LLMs to tabular biomarker data in biomedicine, demonstrating the potential of finetuned table-language models to tackle structured prediction with limited data in biomedical domain. By uniting few-shot LLM reasoning with multimodal biomarker integration, the proposed TAP-GPT paves the way for future LLM-driven multi-agent systems in AD diagnosis and general health informatics.

Our paper makes the following key contributions:

(1) We propose TAP-GPT, a novel framework for few-shot tabular classification on complex, multimodal AD biomarker data that leverages LLM's powerful language ability and encoded table-specific knowledge through our domain-aware adaptation strategy. As far as we are aware, this study represents the first use of LLMs on tabular biomarker datasets, opening new avenues for LLM-powered frameworks.
(2) We demonstrate that TAP-GPT achieves promising performance on the challenging task of few-shot tabular prediction for AD diagnosis using the public QT-PAD benchmark dataset, comparable or even outperforming more advanced generic LLMs and tabular foundation model approaches.
(3) We provide a comprehensive analysis of TAP-GPT that covers different ablations settings and the interpretability analysis of the proposed framework, shedding lights on future development of agentic/multi-agents approach built on tabular LLMs.

We will make our code and framework publicly available to the research community to facilitate further research and application of data-efficient learning in AD diagnosis and other biomedical applications.

2 Methods

We start this section by introducing our few-shot and in-context learning notions **in the context of tabular prediction task** following previous works [9, 10, 23, 24, 29, 31], as they are slightly different from the normal non-tabular scenario with LLMs.

Formally, let our biomarkers dataset be a table $D = \{(x^i, y^i)\}_{i=1}^N$, consisting of N labeled subject samples. Each sample i is represented by a feature vector $x^i = (x_1^i, \ldots, x_d^i) \in \mathbb{R}^d$ containing d biomarkers, and a corresponding clinical diagnosis label $y^i \in \{0, 1\}$. Here, 0 for CN and 1 for AD as in our QT-PAD dataset (more details in the next section). It can be easily generalized to tabular regression or multiclass classification. The table columns, or features, are described by a set of natural language names for biomarkers $F = \{f_1, \ldots, f_d\}$. The goal of our tabular prediction task is to learn a model that, given a new subject's feature vector $x_{test} \in \mathbb{R}^d$ along with contextual examples together in the form of a table, can accurately predict the corresponding label $y_{test} \in \{0, 1\}$. An overview is provided in Figure 1.

In the few-shot setting, the model makes this prediction conditioned on a small set of k labeled examples in the context. This set, $D_{ICL} = \{(x_{ICL}^j, y_{ICL}^j)\}_{j=1}^k$, is provided as rows in a table to guide the model's inference. This tabular ICL process enables the model to approximate $p(y_{test}|x_{test}, D_{ICL})$ without updating its weights during inference [19]. In the paradigm of tabular foundation model TabPFN [9, 10], the training objective for a model with parameters q_θ is to minimize the negative log-likelihood over many such prediction tasks drawn from a training distribution p:

$$\mathcal{L} = \mathbb{E}_{(D_{ICL}, D_{test}) \sim p}\left[-log q_\theta(y_{test}|x_{test}, D_{ICL})\right] \tag{1}$$

where D_{test} is the singleton set to test containing only one element (x_{test}, y_{test}).

Our proposed framework, TAP-GPT, adapts this paradigm with novel modifications by leveraging the specialized architecture of TableGPT2 [23], a multimodal LLM designed for tabular data. To be specific, we follow the above paradigm to construct an input table T of size $m \times n$, where $m = k + 1$ rows and $n = d$ columns, comprising the k in-context examples from D_{ICL} and the single test sample x_{test}. However, the prediction is then generated by the model conditioned on this entire table and a task-specific instruction I, through standard **next token prediction** in a causal LLM:

$$\hat{y}_{test} = \text{TAP-GPT}(T, I) \tag{2}$$

To predict $\hat{y}_{test}$ in Eqn 2, there are methods that rely on simple serialization to convert the data T into a plain text sequence and then directly prompts pretrained LLMs [8, 14]. For example, a serialization component $\texttt{Serialize}(\mathbf{x}^i, y^i, F)$ can take the column names F and convert each row in the table to a textual description like "Thef_1 biomarker of the person is x_1^i. ..."; then a LLM can simply take the serialized text as input to generate the prediction, i.e. $\hat{y}_{test} = \texttt{LLM}(\texttt{Serialize}(\mathbf{x}^i, y^i, F))$. However, such methods lose the structural information in the original table and can also be token-inefficient for large tables [8].

Unlike these methods, our proposed TAP-GPT framework employs TableGPT2's encoder-decoder model architecture with a dedicated semantic table encoder and a LLM decoder. The encoder processes the input table T to generate compact, structure-aware column embeddings $C(T) \in \mathbb{R}^{n \times k' \times d'}$ (where k' is the number of learnable queries and d' is the LLM's embedding dimension), which are then integrated with the textual instruction I and processed by the LLM decoder QWen2.5 [11]. Our TAP-GPT framework is derived by supervised finetuning the decoder of the pretrained TableGPT2 on our specific AD diagnosis task, using the parameter-efficient qLoRA method [4]. The finetuning objective is to minimize the **cross-entropy loss** between the model's prediction $\hat{y}_{train}$ and the true label y_{train} for all constructed training tables, as shown in Figure 1. In the following subsections, we provide more details for the data, framework and experimental setup.

2.1 Datasets

Data used in the preparation of this article is obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database [3] [27, 28]. The ADNI was launched in 2003 as a public-private partnership led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial MRI, PET, other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of mild cognitive impairment (MCI) and early AD. All participants provided written informed consent, and study protocols were approved by each participating site's Institutional Review Board (IRB). Up-to-date information about the ADNI is available at www.adni-info.org.

The data used in this study are a subset of the ADNI 1/Go/2 cohorts, the QT-PAD project data. A QT-PAD is a quantitative template of multimodal biomarkers to describe the progression of AD and is used to predict the presence of AD in patients.

The QT-PAD features include positron emission tomography (PET) measures (FDG PET and Amyloid PET), cerebrospinal fluid measures (CSF ABETA, CSF TAU, CSF PTau), structural MRI-derived measures from FreeSurfer (FS WholeBrain, FS Hippocampus, FS

[3] http://adni.loni.usc.edu

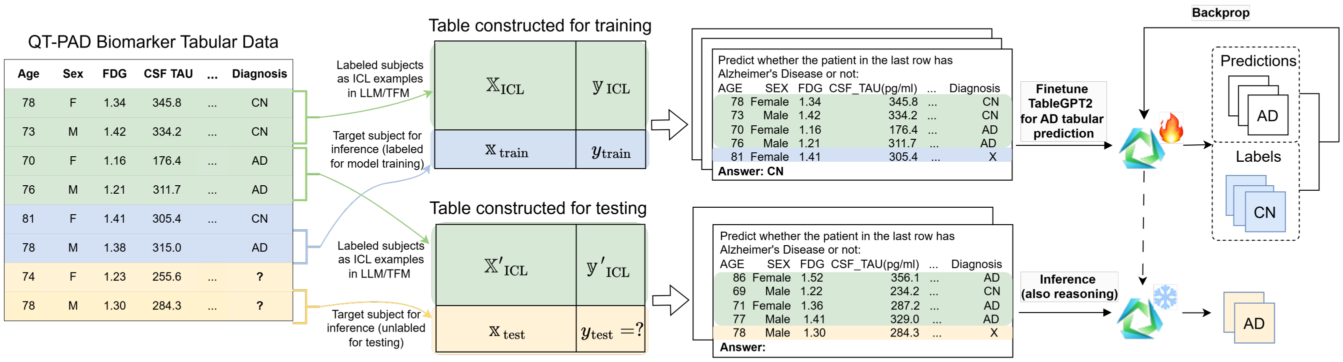

Figure 1: Overview of the TAP-GPT framework. As illustrated in the QT-PAD Biomarker Tabular Data, we split the AD and control subjects into pools for ICL (green color), training (blue color), and testing (yellow color), from which we construct tables used in the downstream tasks (finetuning and inference). CN: cognitively normal. AD: Alzheimer's disease. ICL: in context learning. LLM: large language model. TFM: tabular foundation model.

Entorhinal, FS Ventricles, FD MidTemp, FS Fusiform) and covariates age, APOE4 status, gender, and years of education. Cognitive scores (Alzheimer's Disease Assessment Scale–Cognitive Subscale (ADAS13), Clinical Dementia Rating Sum of Boxes (CDRSB), Mini-Mental State Examination (MMSE), Functional Activities Questionnaire (FAQ), and Rey Auditory Verbal Learning Test learning score (RAVLT.learning)) were excluded because they are highly collinear with diagnostic outcomes. By excluding these scores, we aimed to evaluate model performance on a more challenging and realistic task.

Samples with missing values or any diagnosis besides cognitively normal or AD were excluded, leaving 333 samples with 237 CN and 96 AD patients. All experiments are conducted on this dataset for the binary classification task of CN or AD patients.

To support ICL with tabular LLMs, we create a data splitting strategy using separate pools of data to draw ICL samples for table construction. From the 333 samples, approximately 20% (67 samples) were allocated to the test set, 10% (33 samples) to the validation set, and 40% (125) to the training set. We reserve three non-overlapping pools of approximately 10% (36 samples) for prompt construction: test set ICL pool, validation set ICL pool, and training set ICL pool. The ICL pools are never using in testing, training, or validation directly, they are only used for table construction as ICL examples. These splits are tested with six different seeds for robustness of model evaluation.

We designed four different prompt formats for the binary classification task to test model performance with different contexts and representations of data. Each task format involves generating a prompt with patient-level data and asking the model to predict if a patient is cognitively normal or has AD. The prompt formats vary with two dimensions: context length (zero-shot or few shot), and format (tabular or serialized). Each format is beneath a natural language prompt requesting a classification for that test sample.

(1) **Zero-Shot Tabular** This prompt format contains a single unlabeled row in a tabular input (one target sample). No additional patients are included in the context.
(2) **Zero-Shot Serialized** This prompt format contains a single unlabeled natural language description of a patient's features.
(3) **Few-Shot Tabular** For each unlabeled target sample, k labeled rows are randomly drawn from the corresponding ICL pool as examples. These k rows are displayed in a table above the target sample, which is the final row and is without a label.
(4) **Few-Shot Serialized** Similar to the few-shot tabular format, but each patient's information is serialized into natural language. The k in-context examples have labels and the one target sample does not.

Tabular prompts are displayed with the *head()* function from the pandas library with the target sample in the last row with an 'X' as a label. For serialized prompts, the target sample's diagnosis sentence is omitted. Examples of each prompt format can be found in the Appendix A. These four prompt formats test the impact of exposure to similar patient information (zero-shot vs. few-shot context) and data structure (serialized vs. tabular).

2.2 TAP-GPT framework

TAP-GPT is a domain-adapted tabular LLM repurposes TableGPT2, which is originally optimized for business-intelligence table comprehension tasks, to a completely new structured clinical prediction task for AD. It finetunes TableGPT2 on AD biomarkers data in a structured table format with supervised labels, adapting the model from general tabular understanding to AD-specific tabular prediction. Here, we evaluate if task-specific finetuning for improves performance beyond state-of-the-art methods for tabular few-shot tabular prediction while providing the reasoning capabilities of an LLM. As an overview of the TAP-GPT framework shown in Figure 1, we first extract training and testing samples from the QT-PAD biomarker table, coupling each sample with few-shot examples from the ICL pool. Then, for each sample, we construct a table containing the sample and its ICL examples, and we use these tables to finetune TableGPT2 and subsequently conduct inference (or reasoning) with the resulting model. More details are described in the following paragraphs.

We finetune TableGPT2-7B separately for each prompt format (zero-shot tabular, zero-shot serialized, few-shot tabular, and few-shot serialized) using the qLoRA method. qLoRA is an efficient finetuning approach using low-rank adaptation techniques that enable efficient parameter updates without modifying the full model weights. Finetuning for each task was conducted with the training set and few-shot prompts were generated with the corresponding training ICL pool. Six random seeds (36, 73, 105, 314, 564, and 777) were used for robustness. For each task, the model was finetuned to predict the correct label, Alzheimer's Disease (1) or Cognitively Normal (0). We used a fixed-prompt structure across all samples with the diagnosis label always appearing in the same place after the input.

Finetuning was performed using HuggingFace's transformers and peft libraries with quantized model weights from bitsandbytes (4-bit precision) on each prompt format for each seed. During inference, a custom logits processor constrained the model's output space to only "0" or "1" for the binary classification task.

While we finetuned TableGPT2-7B for each prompt format, the primary focus is the few-shot tabular format to augment TableGPT2 for a tabular prediction task to enable in-context tabular prediction with a limited number of examples. Comparing this setting to serialized prompts and zero-shot contexts isolates the impact of structured data and finetuning on model performance.

In this work, we only finetune TableGPT2's LLM decoder, build on Qwen2.5, keeping the tabular encoder fixed because the available pretrained TableGPT2 only exposes the decoder for downstream adaptation. While this limits the model's capacity to update tabular representations, the encoder's general-purpose table representations can be used while tailoring the LLM with domain knowledge of this clinical prediction task.

2.3 Experimental setup

We designed a series of experiments to evaluate the performance of TAP-GPT and baseline models on a clinical classification task under varying prompt formats and model types, as shown in Figure 2. All experiments were repeated across six random seeds (36, 73, 105, 314, 564, and 777) and evaluation was performed on the test set using the four different prompt formats. The few-shot prompts were generated with k random samples from the test ICL pool, as described in the Data section. All models were evaluated on the same binary classification task using standardized input features and the same test set. Only the prompt format and model type varied across experiments. Performance was primarily measured with AUROC but also evaluated with AUPRC, accuracy, and F1 score.

To select an appropriate number of k in-context examples for the few-shot prompts, we conducted a k-ablation analysis using the validation set. TAP-GPT (finetuned TableGPT2-7B) models were trained separately on the training samples for the tabular few-shot prompt format for each $k \in 2, 4, 6, 8, 10, 12, 16, 20$ across the six seeds. For each configuration, model performance was evaluated with AUROC on the validation set. The k with the highest consistent performance across all seeds on the few-shot tabular task with

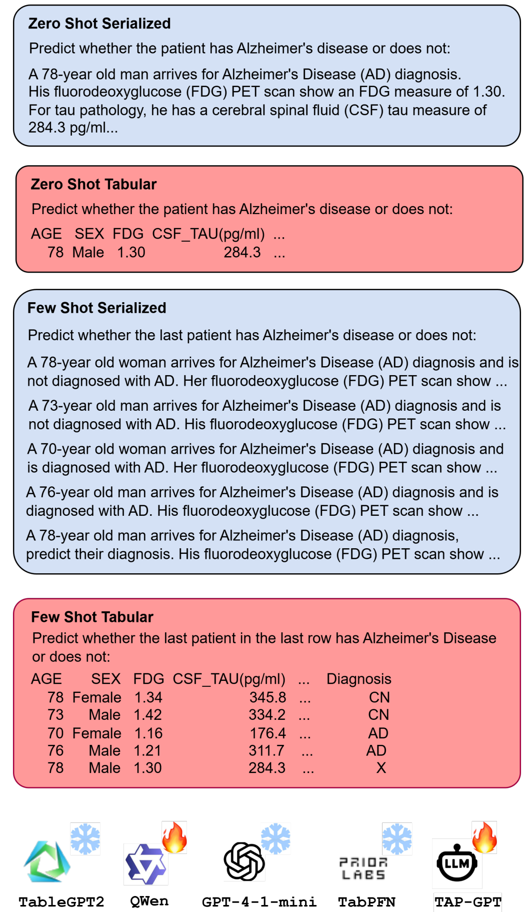

Figure 2: Different prompt formats and models in our experimental setup. The prompts are utilized to feed subject data to the language models, except for TabPFN which directly takes tabular input from CSV files. We show serialized formats in red color and tabular formats in blue, which are used consistently in the subsequent figures as well. For each model, the fire icon denotes model parameters are finetuned, whereas the snowflake icon denotes parameters remain frozen.

Finetuned TableGPT2 was chosen for all few-shot prompts. We also explored larger k values, but those experiments were limited by the GPU memory constraints for larger table sizes.

To benchmark the performance of TAP-GPT, we compared model performance on all four prompt formats across a diverse set of baseline models commonly used for tabular prediction tasks and general-purpose LLMs. The unfinetuned version of TableGPT2 is used to see the impact of task-specific finetuning. To assess the importance of TableGPT's tabular specialization, we tested Qwen2.5-7B-Instruct, the LLM backbone used in TableGPT2, and Qwen3-8B, a more recent general-purpose LLM from the same series. As a sanity check, we included OpenAI's GPT-4-1-mini, a significantly more capable LLM, to benchmark the upper bound of zero- and few-shot performance on this task with general-purpose LLMs. To assess the impact of finetuning TableGPT2's LLM backbone

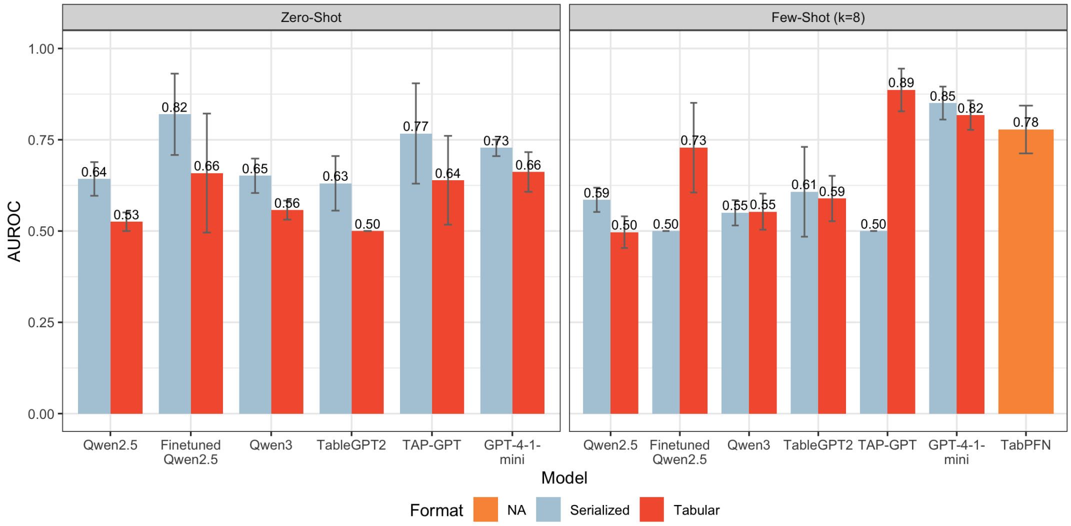

Figure 3: Model performance measured by test AUROC for AD diagnosis prediction across all models for the prompt context zero-shot and few-shot (k = 8). Bars show mean AUROC and error bars show standard deviation across six seeds. TAP-GPT showed the highest performance in the tabular few-shot context (mean AUROC 0.8863 (STD=0.0585)), outperforming the tabular foundation model TabPFN (mean AUROC 0.7781 (STD=0.0654)) and general-purpose LLMs.

without tabular specialization, we finetuned Qwen2.5-7B-Instruct. TabPFN, a state-of-the-art tabular foundation model, was included to represent non-LLM tabular prediction and was tested only on the few-shot task with tabular data.

Hyperparameter tuning was performed using Optuna to maximize AUROC on the validation set. Finetuned TableGPT2-7B models were tuned over key parameters including LoRA rank, dropout rate, learning rate, batch size, weight decay, maximum number of steps, and type of learning rate scheduler. The objective function maximized validation AUROC and the best hyperparameter configurations were chosen for consistent good performance across the six seeds for each model.

All experiments were conducted on a high-performance computing cluster using a single NVIDIA A100 GPU with 80 GB of GPU memory and 160 GB of system memory. Each job was allocated 4 CPU cores and 1 node, with experiments run under a SLURM job scheduler.

3 Results

3.1 Overall Model Performance

TAP-GPT performance was evaluated on the test set across four prompt formats (zero-shot tabular, zero-shot serialized, few-shot tabular, and few-shot serialized) and compared to Qwen2.5-7B-Instruct, Finetuned Qwen2.5-7B-Instruct Qwen3.0-8B, TableGPT2, ChatGPT 4-1-mini, and TabPFN.

As shown in Figure 3 and Table 1, all models performed better in the serialized setting compared to the tabular setting for the zero-shot context, suggesting that models cannot extract meaningful structural information from a single-row input. Finetuned Qwen2.5 shows the strongest performance across all zero-shot contexts in both the serialized (mean AUROC 0.8197 (STD=0.1114)) and tabular settings (mean AUROC 0.6588 (STD=0.1630)). This suggests that finetuning a general-purpose LLM provides sufficient domain knowledge for this clinical prediction task without structural cues from tabular formatting. Finetuning TableGPT also improved performance over its unfinetuned counterpart to a similar extent as Qwen2.5, indicating that in this low-context format, performance gains are likely driven by the underlying LLM backbone rather than tabular prediction specialization.

In the few-shot tabular context shown in Figure 3, TAP-GPT has the strongest performance (mean AUROC 0.8863 (STD=0.0585)), benefiting from pairing the tabular-aware architecture of TableGPT with domain- and task-specific knowledge from finetuning. As expected, TabPFN performs well in this context (mean AUROC 0.7781 (STD=0.0654)), despite being built for larger tables. TAP-GPT performs significantly stronger than TabPFN on the tabular few-shot task (p-value=0.03035). GPT-4-1-mini performs well across both formats in this context, which suggests that updating the backbone of TAP-GPT with more capable base models could further improve prediction performance. Most models underperform in the serialized formatting, often collapsing into single-class predictions, possibly due to prompt complexity or length.

To assess whether TAP-GPT significantly outperforms TabPFN in the few-shot tabular setting, we first tested the normality of the test AUROC differences across six random seeds using the Shapiro-Wilk test. The result (W = 0.899, p = 0.368) does not reject the null hypothesis of normality, justifying the use of a paired t-test. The paired t-test shows that TAP-GPT achieves significantly higher AUROC scores than TabPFN (t = −2.993, p = 0.030), with a mean improvement of 0.108 and a 95% confidence interval of [−0.201, −0.015], indicating consistent performance gains across seeds.

Table 1: AUROC and Accuracy performance metrics for each model for each prompt format. TAP-GPT (bold fonts) demonstrates the strongest performance on the Tabular Few-Shot (k=8) prompt format.

Context	Format	Model	Mean AUROC	AUROC SD	Mean Accuracy	SD Accuracy
Zero-Shot	Serialized	Qwen2.5	0.6428	0.0461	0.6741	0.0447
		Finetuned Qwen2.5	0.8197	0.1114	0.8632	0.0740
		Qwen3	0.6513	0.0471	0.6940	0.0409
		TableGPT2	0.6307	0.0747	0.6492	0.0617
		GPT-4-1-mini	0.7278	0.0226	0.7960	0.0225
		TAP-GPT	**0.7672**	**0.1375**	**0.8258**	**0.0753**
	Tabular	Qwen2.5	0.5255	0.0254	0.7263	0.0154
		Finetuned Qwen2.5	0.6588	0.1630	0.8035	0.0893
		Qwen3	0.5573	0.0260	0.7263	0.0278
		TableGPT2	0.5000	0.0000	0.2836	0.0000
		GPT-4-1-mini	0.6618	0.0544	0.7015	0.0433
		TAP-GPT	**0.6391**	**0.1217**	**0.7562**	**0.0692**
Few-Shot (k=8)	Serialized	Qwen2.5	0.5855	0.0333	0.7288	0.0220
		Finetuned Qwen2.5	0.5000	0.0000	0.7164	0.0000
		Qwen3	0.5500	0.0347	0.6741	0.0476
		TableGPT2	0.6075	0.1230	0.6691	0.0996
		GPT-4-1-mini	0.8505	0.0451	0.8010	0.0548
		TAP-GPT	**0.5000**	**0.0000**	**0.7164**	**0.0000**
	Tabular	Qwen2.5	0.4970	0.0434	0.6741	0.0405
		Finetuned Qwen2.5	0.7284	0.1228	0.8234	0.0690
		Qwen3	0.5532	0.0494	0.7015	0.0490
		TableGPT2	0.5892	0.0621	0.5746	0.0564
		GPT-4-1-mini	0.8174	0.0403	0.7612	0.0481
		TabPFN	0.7781	0.0654	0.8365	0.0493
		TAP-GPT	**0.8863**	**0.0585**	**0.9055**	**0.0488**

3.2 Ablations

3.2.1 K-Ablation Analysis. To determine the optimal k in the few-shot in-context-learning context, we conducted an ablation study on the models TAP-GPT, TableGPT2, and TabPFN for a set of different ks $\{2, 4, 6, 8, 10, 12, 16, 20\}$ (Figure 4). We restricted the upper limit to 20 due to GPU memory constraints. TabPFN showed steadily increasing performance with higher k, consistent with its design for larger tabular input. TableGPT2 performance was relatively steady across k. TAP-GPT showed the highest mean AUROC at $k = 10$ (0.8265, SD=0.1192), but $k = 8$ produced nearly equivalent performance (0.8226, SD=0.05195) with substantially lower variance. For this reason, $k = 8$ was chosen for all few-shot experiments to ensure more consistent performance across random seeds.

3.2.2 Zero-Shot Application to Few-Shot. To evaluate if TableGPT could generalize from a simpler context, zero-shot, to a more complex context, few-shot, we tested the model finetuned on only zero-shot prompts directly on the few-shot prompts. Finetuning on a simpler task could reduce computational complexity and isolate the impact of in-context examples at the time of inference. However, the zero-shot finetuned TableGPT underperformed on the few-shot tabular prompts (mean AUROC of 0.58) compared to TAP-GPT, TableGPT finetuned on few-shot prompts (mean AUROC of 0.88). These results indicate that alignment of prompt formatting for finetuning and inference is critical.

3.3 Interpretability Analysis

A critical requirement for the clinical adoption of any predictive model is interpretability, the ability to understand the rationale behind its decisions. While tabular foundation models like TabPFN demonstrate strong predictive performance, their nature as purely structural foundation models means they cannot provide innate natural language explanations for their predictions. This *black box* characteristic could be a significant barrier in biomedical domains. In contrast, a unique strength of our LLM-based TAP-GPT framework is its capacity to generate human-readable reasoning for its tabular predictions, like AD diagnosis here, a feature crucial for building interpretability and for future integration into trustworthy multi-agent systems (ongoing work).

To explore this capability, we conducted a case study using a minimal Chain-of-Thought (CoT) prompting [26] to explore the reasoning ability of TAP-GPT. By simply modifying the instruction and appending the standard *"Let's think step-by-step"* CoT string to encourage the model's response, we prompted the finetuned TAP-GPT model to output not just a 0/1 prediction, but also the reasoning process that led to it. This was done without providing complex reasoning examples in the prompt, allowing us to assess the model's emergent reasoning abilities after our finetuning on domain-specific, small data.

CoT Reasoning Case Study. We analyzed the model's generated rationales for predictions on the test set. Two examples (manually

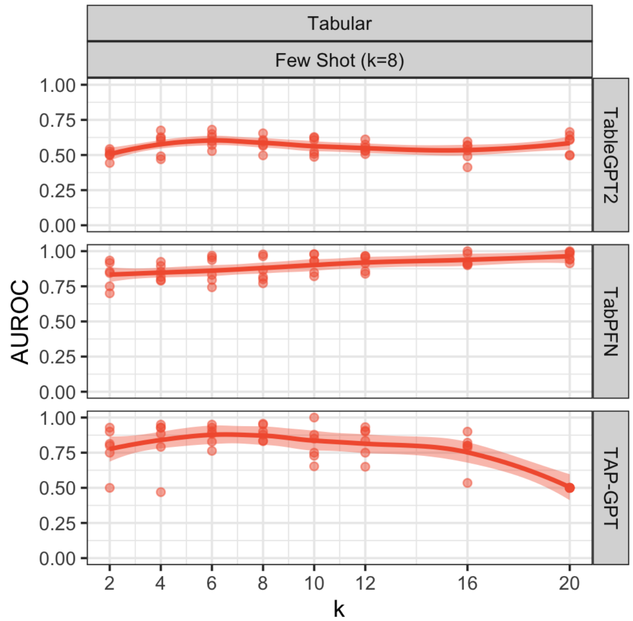

Figure 4: k ablation analysis across TableGPT2, TabPFN, and TAP-GPT. TAP-GPT performs best on $k = 8$ with the least variance across seeds. Each model is evaluated across six seeds on the validation data set.

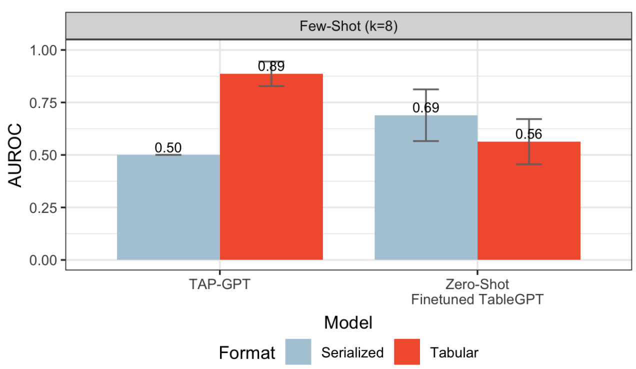

Figure 5: Evaluation of TableGPT2 finetuned on zero-shot prompts used on few-shot prompts. Performance drops significantly, indicating that the finetuned model does not generalize well across prompt formats.

fabricated patient data closely emulating real data) provided in Appendix B illustrate both the CoT prompt and the reasoning process to showcase TAP-GPT's interpretability.

In a representative true positive case (Appendix B, Example 1), TAP-GPT correctly predicted a diagnosis of AD (label 1) and generated a clinically relevant rationale. The model's reasoning correctly identified key risk factors from the patient's data, stating: *"generally, higher values for CSF_ABETA and lower values for CSF_TAU and CSF_PTAU are associated with lower risk of Alzheimer's disease... The patient is male and has APOE4, which increases the risk. The CSF_ABETA level is low (740.4), and the CSF_TAU and CSF_PTAU levels are relatively high (332.0 and 31.58, respectively), indicating a higher risk... Based on the high risk factors (APOE4, low CSF_ABETA, high CSF_TAU and CSF_PTAU), it is likely that this patient has Alzheimer's disease."* This explanation aligns with established AD pathology, where amyloid plaque accumulation in the brain leads to lower CSF Aβ levels, and neuronal injury leads to higher CSF Tau and p-Tau levels. This ability to articulate a step-by-step diagnostic logic based on multiple biomarkers is a significant advantage over non-linguistic models like TabPFN.

In the second case (Appendix B, Example 2), the model also made a correct prediction with step-by-step reasoning. However, by providing the reasoning process, we can spot flaws in the model's understanding, i.e.: *"Patients with Alzheimer's disease (AlzheimersDisease = 1) tend to have higher values for CSF_ABETA, CSF_TAU, and CSF_PTAU."* This statement incorrectly generalizes the elevated pattern for Tau and p-Tau to *Abeta*, contradicting the known biomarker signature of AD. This error suggests that the model, in some cases, may oversimplify patterns from the in-context examples, adopting a naive heuristic that "higher is worse" for all CSF biomarkers. Moreover, this highlights that the reasoning process can reveal the limits of the model and shed light on potential solutions. Maybe the biomedical knowledge encoded in the model is incorrect, or maybe the model is too sensitive to the data distribution of the few-shot examples provided in the prompt. However, a black-box model may never reveal such limits, since the final prediction is correct. The reasoning ability from TAP-GPT allows transparent interpretability analysis through the model's own explanation, providing important insights for further improvements, such as implementing strategies for selecting balanced and representative ICL examples, more finetuning or retrieval argumentation with AD specific materials [15, 32] to revise the knowledge, etc.

In general, although TableGPT2 uses QWen2.5-base-7B as its backbone model, which is not a reasoning model itself, with CoT prompting fed to our finetuned TAP-GPT, we can provide detailed reasoning to the tabular prediction results for AD diagnosis. This is an unique strength over TabPFN and other tabular foundation models (with no language reasoning ability) in particular. It supplies a promising feature for future biomedical tasks and open up possibility for potential multi-agent system design.

4 Discussion

Foundation models for text and images have revolutionized AI research, and there is a growing realization that tabular data, the backbone of clinical and scientific datasets, deserves a similar focus. This work demonstrates the ability of finetuning a tabular LLM model for a clinical prediction task to outperform general-purpose LLMs and a tabular foundation model on a few-shot tabular clinical prediction task. TAP-GPT benefits from the the structured setting with enough examples to reveal structural patterns across rows and columns as well as the domain-specific knowledge obtained from finetuning.

Our work pioneers the application of a tabular LLM to the challenging task of AD diagnosis based on tabular biomarker data in a data-scarce setting. Our proposed TAP-GPT framework successfully adapts TableGPT2 model originally designed for business intelligence into a powerful tool for few-shot AD tabular prediction. The results show that by combining a tabular-aware architecture with domain-specific finetuning, TAP-GPT achieves promising performance for the task. Our in-depth experiments demonstrate the clear synergy between the different components introduced in the TAP-GPT framework. As shown in Figure 3, neither the original TableGPT2 nor the finetuned general-purpose LLM (Qwen2.5-Instruct) could match the performance of TAP-GPT in the crucial few-shot tabular setting. This provides insights that simply having a tabular architecture or domain knowledge alone is insufficient; their integration is likely the key for the improved capability.

An important finding of this study is TAP-GPT's ability to compete with TabPFN in the low-shot realm that is very common in biomedical research. TabPFN, a leading tabular foundation model, represents the new generation of techniques for tabular data, an area deep learning approaches have traditionally struggled with, and introduces a paradigm shift from classic tree-based approaches. Our relatively simple approach performs comparatively with TabPFN in lower-shot contexts, as seen in Figure 4, particularly in the 2-10 range. This suggests that TAP-GPT performs well with interpretability when labeled data are limited, a common context in the biomedical domain. At larger ks, the current framework of TAP-GPT may be limited by context and prompt complexity. TabPFN shows improvement over k, which is expected as it is a more sophisticated, dedicated foundation model built for tabular prediction tasks that can handle up to 10,000 ICL examples [10]. However, it is worth emphasizing that, TAP-GPT, inheriting from TableGPT2, is essentially a multimodal method, connecting a tabular encoder to a large language model decoder. Similar to vision-language models [33], the table and language representations are connected in the latent embedding space through training, allowing TAP-GPT to leverage not only the structural patterns within the table but also the powerful language ability encoded in its LLM backbone. In contrast, TabPFN, although pretrained on millions of tables for few-shot ICL on small to medium-sized tabular data, relies solely on the tabular structure from the input data, without any language modality/ability. Therefore, it does not have the rich semantic knowledge from textual data encoded in LLMs like Qwen that may already have prior knowledge related to AD that can help low-shots predictions, such as familiarity with the biomarker names, etc. This may partially explain why TAP-GPT performs comparably or in some cases even better than TabPFN in the lower-shot settings.

Moreover, the potential for prediction with interpretability results through CoT reasoning provides a unique advantage of TAP-GPT over traditional machine learning models or foundation models like TabPFN. As demonstrated in our interpretability analysis (Section 3.3), with minimal CoT prompting, finetuned TAP-GPT can generate step-by-step rationales for its predictions. Revealing the reasoning that led to a particular conclusion could highlight correct and incorrect assumptions that could be adjusted to improve future predictions for more transparent decision-making in high-stakes domains like clinical prediction. It could also reveal patterns leading to a clinical prediction that have not been prioritized yet. The reasoning we observe arises from general-purpose pretraining and task-specific finetuning, so it is important to understand why the model arrives at a conclusion, whether that is domain-relevant correlations or task-aligned patterns that could be misaligned with domain knowledge. This interpretability directly supports our third contribution and lays the groundwork for more sophisticated, trustworthy AI, such as the multi-agent diagnostic systems we envision for future work. Having said that, the CoT reasoning is an emergent capability from finetuning and can sometimes produce clinically inaccurate statements. We would like to continue work in this area to further understand the reasoning capabilities of TAP-GPT on tabular data to further improve our clinical predictions and provide useful insights.

Even without tabular-specific architecture, OpenAI's GPT-4-1-mini was able to achieve strong performance without finetuning. This indicates the emergent capability of current general-purpose LLMs of tabular understanding and prediction with high-capacity pretraining. Future directions for this framework could include manually decoupling the architecture of TableGPT2 to replace the current LLM decoder, Qwen2.5, with a more modern LLM that already has some tabular understanding capacities. However, the current architecture of TableGPT2 is not open-source, so this limits our ability to modify parts of its structure. Manually recreating TableGPT's architecture, potentially using TabPFN's tabular encoder and a general-purpose LLM like OpenAI's GPT-4-1-mini or Meta's Llama-3-8, could be a future step to further specialize a tabular LLM for clinical prediction tasks.

Our current dataset contains information relating to the ATN (Amyloid, Tau, and Neurodegeneration) framework of AD diagnosis, but we would like to expand to incorporate additional biomarkers like genetic data or imaging data to create a more robust multimodal analysis. Multimodal integration could improve prediction performance but also reveal latent interactions in disease progression and prediction. Systematic ablation or attribution-based analyses could help identify which data types drive model predictions in different contexts, aiding interpretability of results. Additionally, expanding to more difficult tasks like future prediction of AD or differentiation between cognitively normal, mild cognitive impairment, and AD could further distinguish the capabilities of TAP-GPT.

One promising advantage of LLM-based prediction models is their potential for handling missingness. Traditional machine learning methods or TFMs require imputation if missing values occur, which can mislead prediction. Because LLMs operate over text-like serialized inputs, missingness can be implicitly represented by the absence of a value, rather than forcing the model to reason over artificially filled-in data. The LLM could learn context-dependent patterns of missingness and use it to infer as an informative signal, which could further enhance prediction. Accurate handling of missingness is critical in clinical tasks, where data is frequently incomplete or irregular and accurate diagnosis is critical. This is a future area of analysis for TAP-GPT and other general-purpose LLMs on this task.

Lastly, this work lays the foundation for developing LLM-driven multi-agent AD diagnostic systems. We envision a framework where TAP-GPT acts as a specialized biomarker agent, generating some initial diagnosis and rationale. This output could then be passed to another independent neuroimaging agent, or an EHR agent etc., which could leverage additional information and knowledge to cross-validate the clinical reasoning. Such a collaborative system could mitigate the risk of flawed individual judgments and represents a potential step towards building more reliable and trustworthy AI in AD diagnosis and other fields. This work serves as a successful proof-of-concept and provides the foundation for adapting emerging tabular LLMs to solve these critical problems.

5 Conclusion

This work demonstrates the promising strength of TAP-GPT on tabular few-shot prompts for predicting Alzheimer's Disease in patients using clinical biomarkers from the ATN framework. TAP-GPT expands the capabilities of TableGPT2 beyond tabular understanding into tabular prediction. TAP-GPT outperforms TabPFN, a tabular foundation model, and general-purpose LLMs in this low-shot context, leveraging structural information and task-specific finetuning. Beyond performance, TAP-GPT offers interpretability in its reasoning for predictions, which could aid clinical implementation and offer use in multi-agent frameworks. Our results indicate the importance of prompt context (zero-shot vs few shot, tabular vs. serialized) on LLM prediction performance. Our framework serves as a step to apply LLMs to tabular biomarker data to support future clinical prediction tasks and incorporation into multi-agent frameworks.

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A Prompt Formats for AD Prediction

Below, we provide example prompt–output pairs for each prompt format: zero-shot tabular, zero-shot serialized, few-shot tabular, and few-shot serialized. All patient data shown in these examples is entirely fabricated.

Zero-Shot Tabular. **Prompt:**

```
Below is an instruction that describes a task, paired with an input that provides further context. Write
    a response that appropriately completes the request.
### Instruction: Below is a table of patient records. Each column contains features related to Alzheimer'
    s disease. Based on the information, predict whether the patient in the last row has Alzheimer's
    disease (1) or does not (0). Respond only with 1 or 0.
### Input:
AGE GENDER EDUCATION APOE4 FDG AV45 CSF_ABETA(pg/ml) CSF_TAU(pg/ml) CSF_PTAU(pg/ml) WholeBrain
    Hippocampus Entorhinal Ventricles MidTemp Fusiform
 73 Female 11 0 1.131 1.4311 1163.2 305.2 29.48 978382 6728 3278 31733 21383 18321
### Response:
```

Expected Model Output: 0

Zero-Shot Serialized. **Prompt:**

```
Below is an instruction that describes a task, paired with an input that provides further context. Write
    a response that appropriately completes the request.
### Instruction: Below is a table of patient records. Each column contains features related to Alzheimer'
    s disease. Based on the information, predict whether the patient in the last row has Alzheimer's
    disease (1) or does not (0). Respond only with 1 or 0.
### Input:
A 73-year-old woman arrives for Alzheimer's Disease (AD) diagnosis. She has received 11 years of
    education, and she carries 0 copies of the APOE4 genetic variant. She received clinical examinations
    including cerebrospinal fluid (CSF) analysis, positron emission tomography (PET) imaging, and brain
    magnetic resonance imaging (MRI) imaging. For beta-amyloid pathology, she has an amyloid PET measure
    of 1.4311 and a CSF A-beta42 measure of 1163.2 pg/ml. For tau pathology, she has a CSF tau measure of
     305.2 pg/ml and a CSF phosphorylated tau measure of 29.48 pg/ml. For MRI neuroimaging scans, she has
     whole brain volume of 978382, hippocampus region volume of 6728, entorhinal volume of 3278,
    ventricles volume of 31733, middle temporal lobe volume of 21383, and fusiform gyrus volume of 18321.
     In addition, her fluorodeoxyglucose (FDG) PET scan show an FDG measure of 1.131.
### Response:
```

Expected Model Output: 0

Few-Shot Tabular. **Prompt:**

```
Below is an instruction that describes a task, paired with an input that provides further context. Write
    a response that appropriately completes the request.
### Instruction: Below is a table of patient records. Each column contains features related to Alzheimer'
    s disease. The last row is missing a value in the 'AlzheimersDisease' column. Based on the patterns
    in the other rows, predict whether the patient in the last row has Alzheimer's disease (1) or does
    not (0). Respond only with 1 or 0.
### Input:
 AGE GENDER EDUCATION APOE4 FDG AV45 CSF_ABETA(pg/ml) CSF_TAU(pg/ml) CSF_PTAU(pg/ml) WholeBrain
     Hippocampus Entorhinal Ventricles MidTemp Fusiform AlzheimersDisease
  74 Female 12 1 1.4271 0.8999 1821.2 425.8 38.21 9052812 6788 3743 28326 26737 10474 1
  64 Male 14 0 0.8377 0.9868 1555.0 168.2 12.44 1479873 8940 3546 26458 27277 25435 0
  59 Female 20 1 1.9326 1.4429 637.0 280.2 18.33 1043476 9374 4344 36277 13452 15366 0
  70 Female 18 2 1.6361 1.3294 1113.4 400.6 23.73 1135319 8747 3967 27978 15263 15436 0
  73 Female 12 1 1.0631 1.0242 1800.0 502.2 48.13 988354 7252 3700 20665 16543 18425 1
  60 Male 14 1 1.2367 1.1111 992.2 183.8 12.70 1242152 8630 4355 45372 35421 26453 0
  75 Male 12 0 1.3745 1.8312 721.4 502.2 53.37 1006782 5834 3005 48327 14325 24352 1
  73 Female 9 2 0.9327 1.8421 1004.2 487.6 48.42 893277 5032 2477 27652 15234 10043 1
  73 Female 11 0 1.1313 1.4311 1163.2 305.2 29.48 978382 6728 3278 31733 21383 18321 X
### Response:
```

Expected Model Output: 0

Few-Shot Serialized. **Prompt:**

```
Below is an instruction that describes a task, paired with an input that provides further context. Write
    a response that appropriately completes the request.
### Instruction: Below is a serialization of patient records. Each record contains features related to
    Alzheimer's disease. The last patient has a missing Alzheimer's diagnosis. Based on the patterns in
    the other records, predict whether the patient in the last record has Alzheimer's disease (1) or does
     not (0). Respond only with 1 or 0.
### Input:
A 74-year-old woman arrives for Alzheimer's Disease (AD) diagnosis and is diagnosed with Alzheimer's
    disease. She has received 12 years of education, and she carries 1 copy of the APOE4 genetic variant.
     She received clinical examinations including cerebrospinal fluid (CSF) analysis, positron emission
    tomography (PET) imaging, and brain magnetic resonance imaging (MRI) imaging. For beta-amyloid
    pathology, she has an amyloid PET measure of 0.8999 and a CSF A-beta42 measure of 1821.2 pg/ml. For
    tau pathology, she has a CSF tau measure of 425.8 pg/ml and a CSF phosphorylated tau measure of 38.21
     pg/ml. For MRI neuroimaging scans, she has whole brain volume of 9052812, hippocampus region volume
    of 6788, entorhinal volume of 3743, ventricles volume of 28326, middle temporal lobe volume of 26737,
     and fusiform gyrus volume of 10474. In addition, her fluorodeoxyglucose (FDG) PET scan shows an FDG
    measure of 1.4271.
A 64-year-old man arrives for Alzheimer's Disease (AD) diagnosis and is not diagnosed with Alzheimer's
    disease. He has received 14 years of education, and he carries 0 copies of the APOE4 genetic variant.
     He received clinical examinations including cerebrospinal fluid (CSF) analysis, positron emission
    tomography (PET) imaging, and brain magnetic resonance imaging (MRI) imaging. For beta-amyloid
    pathology, he has an amyloid PET measure of 0.9868 and a CSF A-beta42 measure of 1555.0 pg/ml. For
    tau pathology, he has a CSF tau measure of 168.2 pg/ml and a CSF phosphorylated tau measure of 12.44
    pg/ml. For MRI neuroimaging scans, he has whole brain volume of 1479873, hippocampus region volume of
     8940, entorhinal volume of 3546, ventricles volume of 26458, middle temporal lobe volume of 27277,
    and fusiform gyrus volume of 25435. In addition, his fluorodeoxyglucose (FDG) PET scan shows an FDG
    measure of 0.8377.
A 59-year-old woman arrives for Alzheimer's Disease (AD) diagnosis and is not diagnosed with Alzheimer's
    disease. She has received 20 years of education, and she carries 1 copy of the APOE4 genetic variant.
     She received clinical examinations including cerebrospinal fluid (CSF) analysis, positron emission
    tomography (PET) imaging, and brain magnetic resonance imaging (MRI) imaging. For beta-amyloid
    pathology, she has an amyloid PET measure of 1.4429 and a CSF A-beta42 measure of 637.0 pg/ml. For
    tau pathology, she has a CSF tau measure of 280.2 pg/ml and a CSF phosphorylated tau measure of 18.33
     pg/ml. For MRI neuroimaging scans, she has whole brain volume of 1043476, hippocampus region volume
    of 9374, entorhinal volume of 4344, ventricles volume of 36277, middle temporal lobe volume of 13452,
     and fusiform gyrus volume of 15366. In addition, her fluorodeoxyglucose (FDG) PET scan shows an FDG
    measure of 1.9326.
A 70-year-old woman arrives for Alzheimer's Disease (AD) diagnosis and is not diagnosed with Alzheimer's
    disease. She has received 18 years of education, and she carries 2 copies of the APOE4 genetic
    variant. She received clinical examinations including cerebrospinal fluid (CSF) analysis, positron
    emission tomography (PET) imaging, and brain magnetic resonance imaging (MRI) imaging. For beta-
    amyloid pathology, she has an amyloid PET measure of 1.3294 and a CSF A-beta42 measure of 1113.4 pg/
    ml. For tau pathology, she has a CSF tau measure of 400.6 pg/ml and a CSF phosphorylated tau measure
    of 23.73 pg/ml. For MRI neuroimaging scans, she has whole brain volume of 1135319, hippocampus region
     volume of 8747, entorhinal volume of 3967, ventricles volume of 27978, middle temporal lobe volume
    of 15263, and fusiform gyrus volume of 15436. In addition, her fluorodeoxyglucose (FDG) PET scan
    shows an FDG measure of 1.6361.
A 73-year-old woman arrives for Alzheimer's Disease (AD) diagnosis and is diagnosed with Alzheimer's
    disease. She has received 12 years of education, and she carries 1 copy of the APOE4 genetic variant.
     She received clinical examinations including cerebrospinal fluid (CSF) analysis, positron emission
    tomography (PET) imaging, and brain magnetic resonance imaging (MRI) imaging. For beta-amyloid
    pathology, she has an amyloid PET measure of 1.0242 and a CSF A-beta42 measure of 1800.0 pg/ml. For
    tau pathology, she has a CSF tau measure of 502.2 pg/ml and a CSF phosphorylated tau measure of 48.13
     pg/ml. For MRI neuroimaging scans, she has whole brain volume of 988354, hippocampus region volume
    of 7252, entorhinal volume of 3700, ventricles volume of 20665, middle temporal lobe volume of 16543,
     and fusiform gyrus volume of 18425. In addition, her fluorodeoxyglucose (FDG) PET scan shows an FDG
    measure of 1.0631.
```

```
A 60-year-old man arrives for Alzheimers Disease (AD) diagnosis and is not diagnosed with Alzheimer's
    disease. He has received 14 years of education, and he carries 1 copy of the APOE4 genetic variant.
    He received clinical examinations including cerebrospinal fluid (CSF) analysis, positron emission
    tomography (PET) imaging, and brain magnetic resonance imaging (MRI) imaging. For beta-amyloid
    pathology, he has an amyloid PET measure of 1.1111 and a CSF A-beta42 measure of 992.2 pg/ml. For tau
     pathology, he has a CSF tau measure of 183.8 pg/ml and a CSF phosphorylated tau measure of 12.70 pg/
    ml. For MRI neuroimaging scans, he has whole brain volume of 1242152, hippocampus region volume of
    8630, entorhinal volume of 4355, ventricles volume of 45372, middle temporal lobe volume of 35421,
    and fusiform gyrus volume of 26453. In addition, his fluorodeoxyglucose (FDG) PET scan shows an FDG
    measure of 1.2367.
A 75-year-old man arrives for Alzheimer's Disease (AD) diagnosis and is diagnosed with Alzheimer's
    disease. He has received 12 years of education, and he carries 0 copies of the APOE4 genetic variant.
     He received clinical examinations including cerebrospinal fluid (CSF) analysis, positron emission
    tomography (PET) imaging, and brain magnetic resonance imaging (MRI) imaging. For beta-amyloid
    pathology, he has an amyloid PET measure of 1.8312 and a CSF A-beta42 measure of 721.4 pg/ml. For tau
     pathology, he has a CSF tau measure of 502.2 pg/ml and a CSF phosphorylated tau measure of 53.37 pg/
    ml. For MRI neuroimaging scans, he has whole brain volume of 1006782, hippocampus region volume of
    5834, entorhinal volume of 3005, ventricles volume of 48327, middle temporal lobe volume of 14325,
    and fusiform gyrus volume of 24352. In addition, his fluorodeoxyglucose (FDG) PET scan shows an FDG
    measure of 1.3745.
A 73-year-old woman arrives for Alzheimer's Disease (AD) diagnosis, predict their diagnosis. She has
    received 9 years of education, and she carries 2 copies of the APOE4 genetic variant. She received
    clinical examinations including cerebrospinal fluid (CSF) analysis, positron emission tomography (PET
    ) imaging, and brain magnetic resonance imaging (MRI) imaging. For beta-amyloid pathology, she has an
     amyloid PET measure of 1.8421 and a CSF A-beta42 measure of 1004.2 pg/ml. For tau pathology, she has
     a CSF tau measure of 487.6 pg/ml and a CSF phosphorylated tau measure of 48.42 pg/ml. For MRI
    neuroimaging scans, she has whole brain volume of 893277, hippocampus region volume of 5032,
    entorhinal volume of 2477, ventricles volume of 27652, middle temporal lobe volume of 15234, and
    fusiform gyrus volume of 10043. In addition, her fluorodeoxyglucose (FDG) PET scan shows an FDG
    measure of 0.9327.
### Response:
```

Expected Model Output: 0

B Details of Reasoning Analysis

Here, we present the reasoning prompt used in our interpretability analysis of TAP-GPT, designed with minimal CoT to limit confoundings, along with two representative examples (few-shot ICL k=8, 16) for tabular AD prediction. All patient data shown in these examples is entirely fabricated.

Simple CoT Reasoning added to Few-Shot Tabular case. **Prompt:**

```
Below is an instruction that describes a task, paired with an input that provides further context. Write
    a response that appropriately completes the request.
### Instruction: Below is a table of patient records. Each column contains features related to Alzheimer'
    s disease. The last row is missing a value in the 'AlzheimersDisease' column. Based on the patterns
    in the other rows, predict whether the patient in the last row has Alzheimer's disease (1) or does
    not (0). Respond with your reasoning and the prediction answer (1 or 0).
### Input: <Same format as the Few-Shot Tabular case in Section A>
### Response: Let's think step-by-step
```

Example 1 Output with Reasoning (Correct). **K=8:**

```
To predict whether the patient in the last row has Alzheimer's disease, we need to analyze the patterns
    in the other rows.

1. **AGE**: The ages range from 61 to 79 years.
2. **GENDER**: Predominantly female, but there are also males.
3. **EDUCATION**: Education levels vary from 8 to 20 years.
4. **APOE4**: Some patients have APOE4 (1), while others do not (0).
5. **FDG**, **AV45**, **CSF_ABETA**, **CSF_TAU**, **CSF_PTAU**, **WholeBrain**, **Hippocampus**, **
    Entorhinal**, **Ventricles**, **MidTemp**, **Fusiform**: These values vary across patients, but
    generally, higher values for CSF_ABETA and lower values for CSF_TAU and CSF_PTAU are associated with
    lower risk of Alzheimer's disease.
```

```
Given the last row:
<Details ignored here>

**Reasoning**:
- The patient is male and has APOE4, which increases the risk.
- The CSF_ABETA level is low (740.4), and the CSF_TAU and CSF_PTAU levels are relatively high (332.0 and
    31.58, respectively), indicating a higher risk.
- Other biomarkers like FDG and AV45 are within a range that might suggest some cognitive impairment but
    not necessarily Alzheimer's disease.

**Prediction**:
Based on the high risk factors (APOE4, low CSF_ABETA, high CSF_TAU and CSF_PTAU), it is likely that this
    patient has Alzheimer's disease.

**Answer: 1**
```

True Label: 1

Example 2 Output with Reasoning (Flawed). **K=16:**

```
To predict whether the patient in the last row has Alzheimer's disease (1) or does not (0), we need to
    analyze the patterns in the other rows. Here are the steps:

1. **Age**: The age range is from 65 to 87 years.
2. **Gender**: Predominantly female, but there are also males.
3. **Education**: Varies from 10 to 20 years.
4. **APOE4**: Most patients have APOE4 (1), but some do not (0).
5. **FDG**: Values range from 0.9259 to 1.5015.
6. **AV45**: Values range from 1.0074 to 1.5009.
7. **CSF_ABETA**: Values range from 466.2 to 1701.0 pg/ml.
8. **CSF_TAU**: Values range from 10.85 to 421.6 pg/ml.
9. **CSF_PTAU**: Values range from 10.85 to 42.01 pg/ml.
<More details ignored here>

**Observations**:
- Patients with Alzheimer's disease (AlzheimersDisease = 1) tend to have higher values for CSF_ABETA,
    CSF_TAU, and CSF_PTAU.
- Patients without Alzheimer's disease (AlzheimersDisease = 0) tend to have lower values for these
    biomarkers.
- The presence of APOE4 (APOE4 = 1) is common among patients with Alzheimer's disease.

**Prediction**:
Given the last row's data:
- CSF_ABETA: 465.8 pg/ml (lower than many with Alzheimer's)
- CSF_TAU: 247.2 pg/ml (lower than many with Alzheimer's)
- CSF_PTAU: 25.07 pg/ml (lower than many with Alzheimer's)
- APOE4: 0 (not present)

Based on these observations, the patient in the last row is less likely to have Alzheimer's disease.

**Prediction Answer**: 0
```

True Label: 0

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